

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

2026-04-26

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1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions
- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude

is

$$A_{\text{influence}}(\text{condition}) = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|,$$

extracted via **emtrends**. Slot mean differences are retained as a supplemental view. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	38	342
human-ai	100	851
human-human	36	324

3 Preprocessing

Slot-level rows: 2934

Metric rows: 8802

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from resonance $\sim$ surprise *
slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	0.987	0.887	-0.099	0.036	-2.777	0.006	0.099	**
Human-Human	0.890	0.906	0.016	0.049	0.336	0.737	0.016	
AI-AI	0.995	1.018	0.023	0.042	0.547	0.584	0.023	

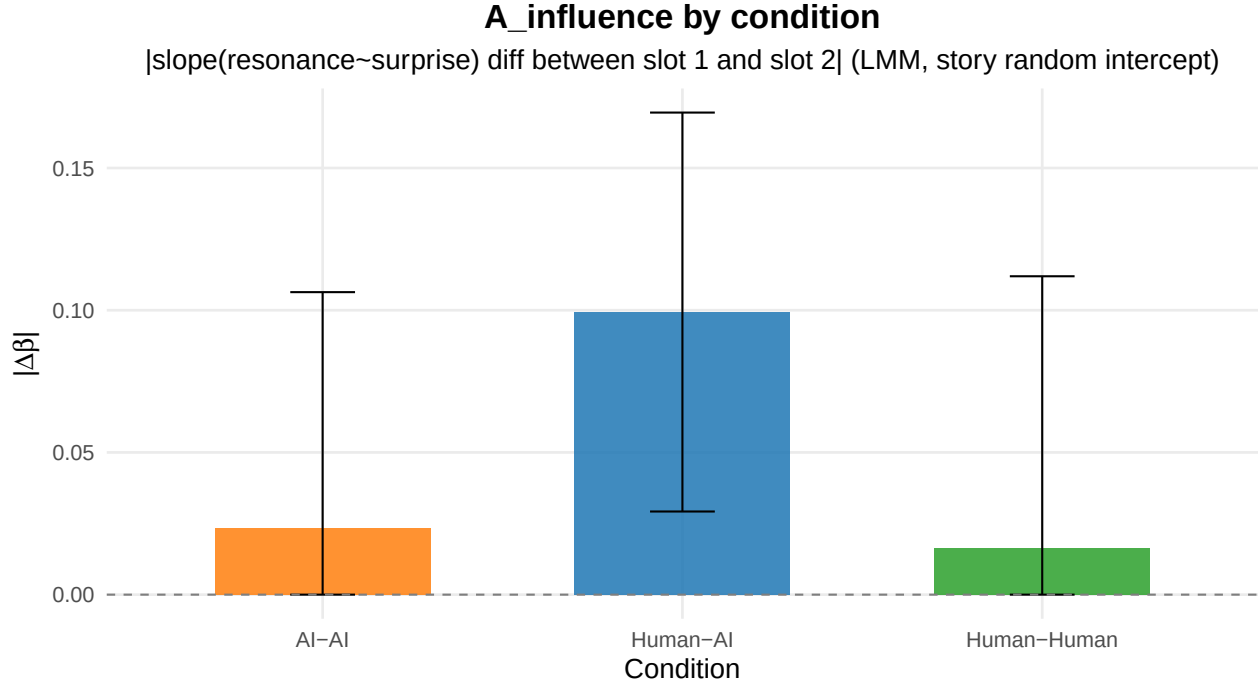


Table 3: Robustness: per-story $|\text{slope}_1 - \text{slope}_2|$ (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	100	0.305	0.370
Human-Human	36	0.195	0.313
AI-AI	38	0.192	0.275

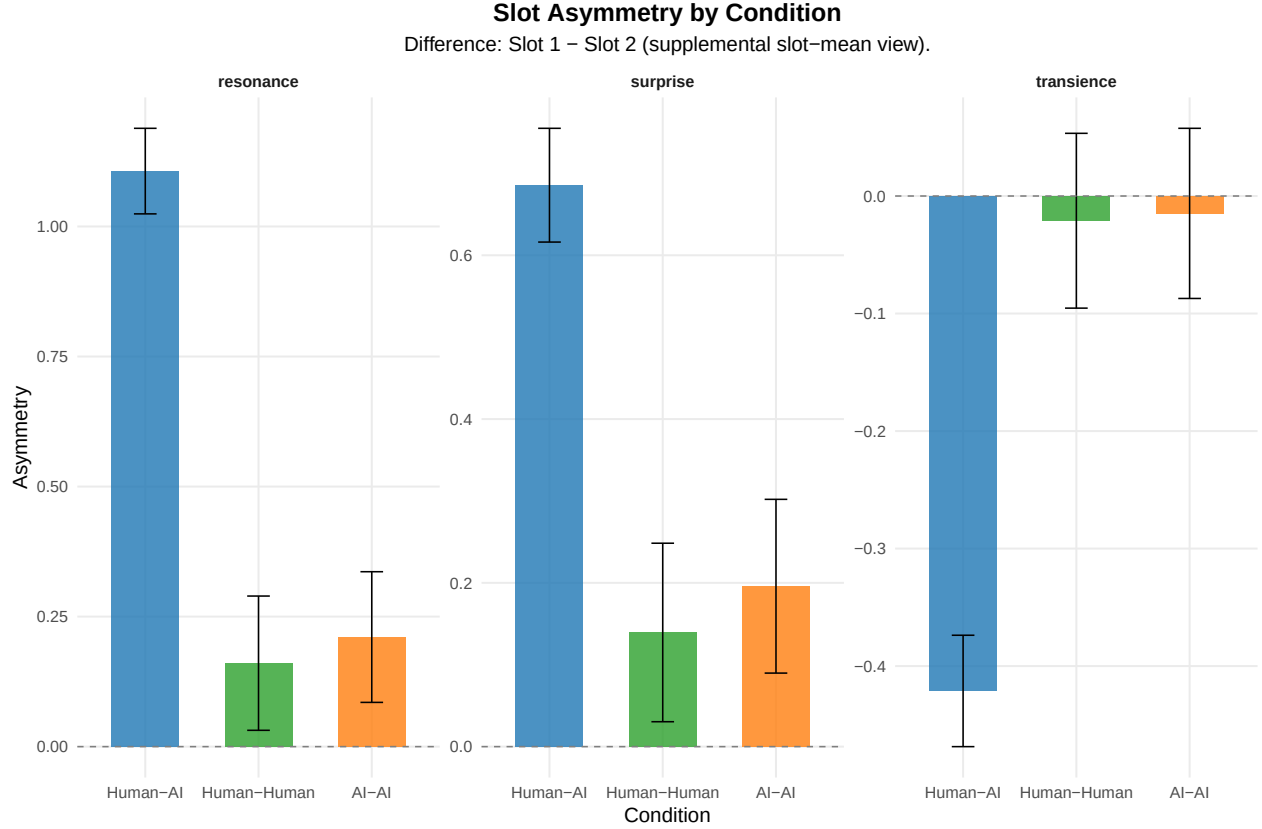
```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 4.8983, df = 2, p-value = 0.08637
```

5 Supplemental: Slot Mean Comparison

Table 4: Slot asymmetry by condition for novelty metrics

metric	condition	slot_1_mean	slot_2_mean	difference	se	t	p	sig
surprise	Human-AI	-1.312	-1.998	0.686	0.035	19.348	0.000	***
surprise	Human-Human	-1.256	-1.396	0.139	0.056	2.507	0.012	*
surprise	AI-AI	-1.661	-1.857	0.196	0.054	3.619	0.000	***
transience	Human-AI	-1.035	-0.614	-0.421	0.024	-17.417	0.000	***
transience	Human-Human	-0.584	-0.563	-0.021	0.038	-0.554	0.579	
transience	AI-AI	-0.913	-0.898	-0.015	0.037	-0.400	0.689	

metric	condition	slot_1_mean	slot_2_mean	difference	se	t	p	sig
resonance	Human-AI	-0.275	-1.381	1.106	0.042	26.370	0.000	***
resonance	Human-Human	-0.672	-0.833	0.160	0.066	2.437	0.015	*
resonance	AI-AI	-0.749	-0.959	0.211	0.064	3.287	0.001	**

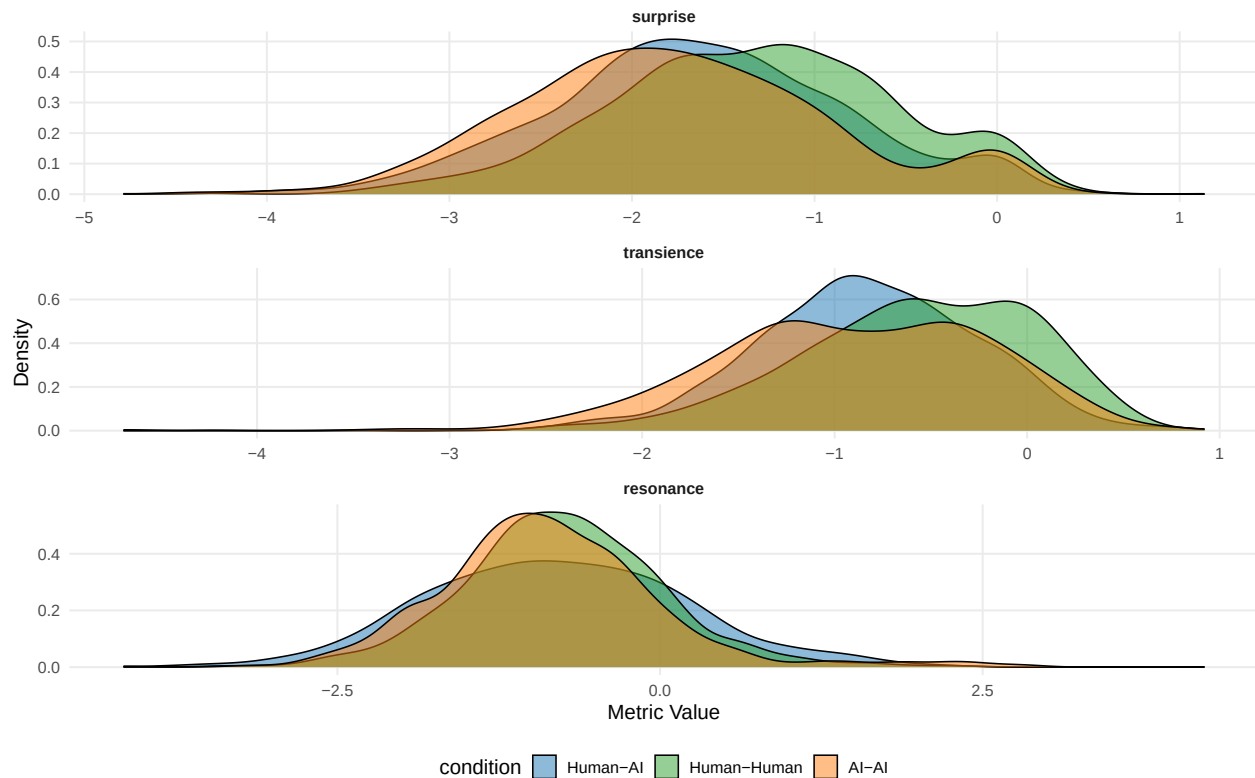


6 Supplemental: General Condition-Level Comparison

Table 5: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	100	1602	-1.631	0.812
Human-AI	transience	100	1602	-0.838	0.578
Human-AI	resonance	100	1602	-0.794	1.015
Human-Human	surprise	36	648	-1.326	0.773
Human-Human	transience	36	648	-0.573	0.626
Human-Human	resonance	36	648	-0.753	0.782
AI-AI	surprise	38	684	-1.759	0.845
AI-AI	transience	38	684	-0.905	0.700
AI-AI	resonance	38	684	-0.854	0.866

Novelty Metric Distributions by Condition



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6916.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5986 -0.6173 -0.0533  0.5909  3.3130
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 0.08696  0.2949
## Residual                0.57179  0.7562
## Number of obs: 2934, groups: conversation_id, 174
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -1.63356    0.03504 178.33980 -46.624 < 2e-16 ***
## conditionHuman-Human    0.30764    0.06727 169.92530  4.573 9.23e-06 ***
## conditionAI-AI    -0.12570    0.06597 170.04592 -1.905  0.0584 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```

##          (Intr) cndH-H
## cndtnHmn-Hm -0.521
## condtnAI-AI -0.531  0.277
## contrast          estimate      SE  df t.ratio p.value
## (Human-AI) - (Human-Human)  -0.308 0.0673 168  -4.573  <.0001
## (Human-AI) - (AI-AI)         0.126 0.0660 168   1.905  0.1753
## (Human-Human) - (AI-AI)      0.433 0.0801 165   5.407  <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4749.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.9968 -0.5815  0.0479  0.6379  2.7175
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1225   0.3499
## Residual                0.2587   0.5086
## Number of obs: 2934, groups: conversation_id, 174
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.83797    0.03724 174.42602 -22.503  < 2e-16 ***
## conditionHuman-Human  0.26466    0.07202 171.06089   3.675  0.000318 ***
## conditionAI-AI     -0.06720    0.07062 171.10944  -0.952  0.342672
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) cndH-H
## cndtnHmn-Hm -0.517
## condtnAI-AI -0.527  0.273
## contrast          estimate      SE  df t.ratio p.value
## (Human-AI) - (Human-Human)  -0.2647 0.0720 170  -3.675  0.0010
## (Human-AI) - (AI-AI)         0.0672 0.0706 170   0.952  1.0000
## (Human-Human) - (AI-AI)      0.3319 0.0860 168   3.857  0.0005
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)

```

```
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 7937.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6466 -0.6415 -0.0268  0.6083  5.3667
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0000    0.000
## Residual                0.8723    0.934
## Number of obs: 2934, groups: conversation_id, 174
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.79351    0.02333 2931.00000  -34.006  <2e-16 ***
## conditionHuman-Human    0.04090    0.04348 2931.00000   0.941   0.347
## conditionAI-AI        -0.06058    0.04266 2931.00000  -1.420   0.156
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.537
## condtnAI-AI -0.547  0.294
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
## contrast              estimate      SE df t.ratio p.value
## (Human-AI) - (Human-Human) -0.0409 0.0435 160  -0.941  1.0000
## (Human-AI) - (AI-AI)         0.0606 0.0427 161   1.420  0.4726
## (Human-Human) - (AI-AI)       0.1015 0.0512 151   1.982  0.1479
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

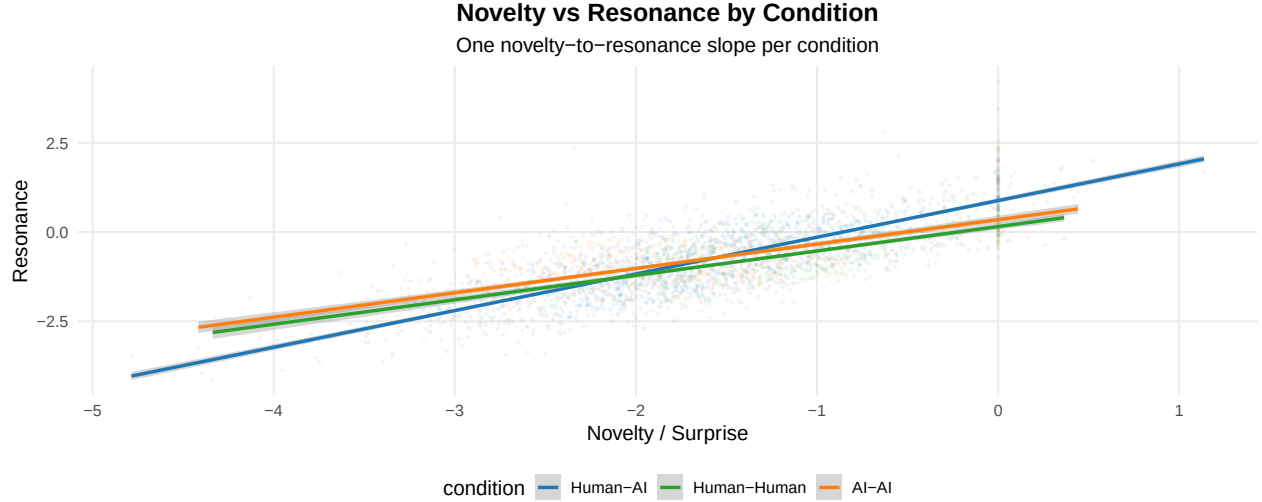
7 Supplemental: Pooled Novelty-to-Resonance Relationship

```
## Novelty-to-resonance model:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
## Data: df_slots
##
## REML criterion at convergence: 4721.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.7255 -0.6338 -0.0508  0.5874  6.9187
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0000    0.000
## Residual                0.8723    0.934
## Number of obs: 2934, groups: conversation_id, 174
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.79351    0.02333 2931.00000  -34.006  <2e-16 ***
## conditionHuman-Human    0.04090    0.04348 2931.00000   0.941   0.347
## conditionAI-AI        -0.06058    0.04266 2931.00000  -1.420   0.156
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.537
## condtnAI-AI -0.547  0.294
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
## contrast              estimate      SE df t.ratio p.value
## (Human-AI) - (Human-Human) -0.0409 0.0435 160  -0.941  1.0000
## (Human-AI) - (AI-AI)         0.0606 0.0427 161   1.420  0.4726
## (Human-Human) - (AI-AI)       0.1015 0.0512 151   1.982  0.1479
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

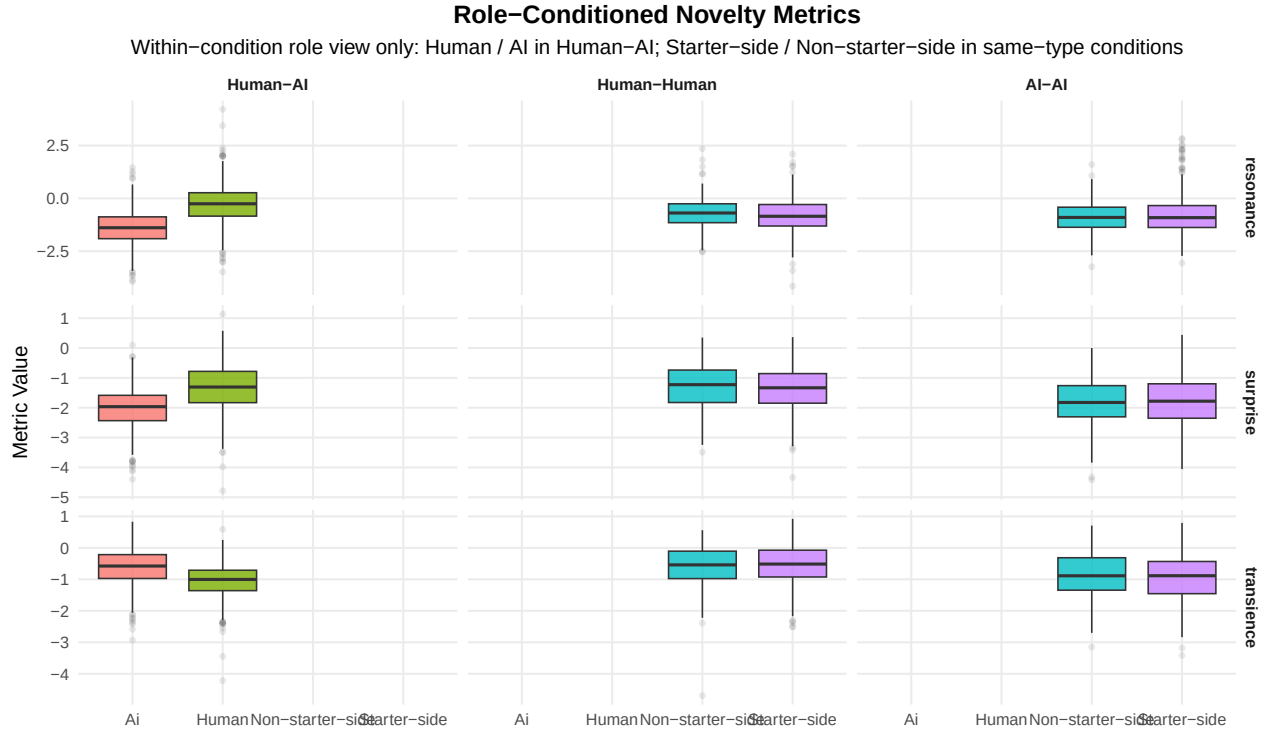
```

## conversation_id (Intercept) 0.1176 0.3430
## Residual 0.2553 0.5053
## Number of obs: 2934, groups: conversation_id, 174
##
## Fixed effects:
##
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 0.98477 0.04526 375.59879 21.760 < 2e-16
## surprise 1.08983 0.01633 2821.06304 66.753 < 2e-16
## conditionHuman-Human -0.55643 0.08487 324.33578 -6.556 2.18e-10
## conditionAI-AI -0.12703 0.08783 392.09960 -1.446 0.148885
## surprise:conditionHuman-Human -0.19917 0.03341 2883.30909 -5.962 2.80e-09
## surprise:conditionAI-AI -0.11680 0.03125 2898.59138 -3.738 0.000189
##
## (Intercept) ***
## surprise ***
## conditionHuman-Human ***
## conditionAI-AI
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise 0.590
## cndtnHmn-Hm -0.533 -0.314
## condtnAI-AI -0.515 -0.304 0.275
## srprs:cnH-H -0.288 -0.489 0.551 0.148
## srprs:AI-AI -0.308 -0.522 0.164 0.614 0.255
##
## Condition-specific novelty-to-resonance slopes:
## condition surprise.trend SE df lower.CL upper.CL
## Human-AI 1.090 0.0163 2826 1.058 1.122
## Human-Human 0.891 0.0292 2899 0.833 0.948
## AI-AI 0.973 0.0267 2916 0.921 1.025
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

8 Supplemental: Role-Level View



9 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from `resonance ~ surprise * slot + (1|conversation_id)`. Per-story OLS slope differences are reported as a robustness layer.

Supplemental views retained:

- slot mean differences per metric (LMM `value ~ slot * condition + (1|story)`)
- condition-level marginal distributions and pooled condition LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction

- within-condition role box-plots (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type conditions)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)
- Human-AI decomposition (`speaker_type` \u00d7 `speaker_is_starter` within Human-AI only)
- length-controlled variant (+ `nwords`) mirroring the paper

Analysis completed: 2026-04-26 22:09:55.858893